QP CODE - 25900323

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION OCTOBER 2025

THIRD SEMESTER

Discipline Specific Core Course (DSC) – MG3DSCCSC201 – DATA STRUCTURES USING C

(2024 ADMISSION ONWARDS)

Duration : 1 Hour 30 Minutes Max Marks : 50

Remember(K), Understand(U), Apply(A), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overll outcome attainability.

ANSWER KEY

Part A (Very Short Answer Questions)

Answer all questions – Each question carries 2 marks

1. Define linear search. [K]/[CO1]
   Linear search is a searching method in which each element of the list is checked one by one from the beginning until the desired element is found or the end of the list.
2. List any two basic operations of a stack. [K]/[CO2]
   Push – inserting an element to the stack (1 mark)
   Pop – deleting an element from the stack (1 mark)
3. Write any two uses of a circular linked list. [K]/[CO3]
   (implement circular queue, Playing music playlist, Buffer management, Round Robin Scheduling, etc.)
   (For correct 2 uses give 2 marks)
4. Define the use of base condition in recursion. [K]/[CO1]
   It is the termination point in recursion prevent it from calling itself endlessly.
5. What is the condition for inserting a node in the left subtree of a BST? [K]/[CO3]
   All nodes in the left subtree of a BST should be less than the values of the parent node.

[2*5=10]

Part B (Short Answer Questions)

Answer any 5 questions – Each question carries 5 marks

6. Write the algorithm for merge sort. [K]/[CO1]
   (If the algorithm is correct, give full marks; otherwise, award marks accordingly.)
7. Convert the infix expression A+B/C-D to prefix. [A]/[CO2]

(In the syllabus only infix to postfix conversion is given. If this question is attempted, please award full marks)

8. Differentiate between binary tree and binary search tree in terms of structure and operations. [An]/[CO3]

   BT – A hierarchical tree where each node has at most two children.

   BST – A special BT where left child value less than the parent node and right child value greater than the parent nod.

   (If explained correctly with neat figure representation give full marks, otherwise award marks accordingly)

9. List the difference between traversal and searching operations in data structures. [K]/[CO1]

   Traversal – visit all elements exactly once.

   Searching - search for a desired element, not compulsory to visit all elements.

   (minimum 3 difference – give full marks)

10. Describe the array representation of a queue. How are the 'front' and 'rear' pointers managed. [U]/[CO2]

    (Definition - 2 marks, figure – 1 marks, explanation – 2 marks)

11. Compare ancestors, descendants and siblings of a node in a tree structure. [An]/[CO3]

    Ancestors – nodes that lie on the path from root to that node excluding the node itself.

    Descendants – nodes that lie below the given node in the tree.

    Siblings – nodes that share the same parent.

    (Explain with example give 5 marks)

12. State the different parameter passing methods in functions. [K]/[CO1]

    (Call by value and call by reference – explanation with example give 5 marks)

[5*5=25]

Part C – Essay Questions

Answer any 1 question – Each question carries 15 marks

13. Discuss in detail the different operations on arrays with suitable examples. [U]/[CO1]

    (Definition of array – 2 marks, figure – 1 marks, explain any 4 operations with examples – 3 marks each)

14. Explain all operations of circular queue with the help of diagram. [U]/[CO2]

    (Definition – 2 marks, figure – 1 mark, explain operations with neat diagram – 6 marks each)

[15*1=15]